

RetailRadar

Detecting the **start** and the **end** of retail euphoria

A walk-forward study on nine years of social-media attention

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Full 77-page version: `retailradar_presentation.pdf`

The job, and the one rule behind it

What a PM asked for

- Tell me when retail is getting *euphoric* about something I hold.
- Tell me when that euphoria is *fading* — before the price tells me.
- Speak **rarely**. A light that is always on is not a light.

So success was defined before any test was run:

an alert inside [peak – 30d, peak + 1d] of a genuine top, where a top is a run of $\geq 25\%$ (ETF) / $\geq 50\%$ (single) followed by a $\geq 15\%$ / $\geq 30\%$ drawdown within 90 days.

SO WHAT The deliverable is a *risk* instrument — a warning light for crowded longs — not an alpha signal. Everything after this slide is tested against that claim and nothing wider.

The one rule the project is built on

Every number in this system is one of:

- **learned** from past data only, then frozen;
- **derived** from a stated budget or definition;
- a **convention**, published with the ablation that shows what it costs;
- a **desk decision**, named as one.

There is no fourth category. “It looked about right” is not in the register.

What is actually measured — and what never is

Euphoria = the crowd has stopped analysing a name and started celebrating it. Four crowd measurements, each ranked against *that name's own* trailing year:

1. **Attention extremity** — you cannot be euphoric quietly.
2. **Sustained bullishness** — weeks of one-way lean, not one loud afternoon.
3. **Crowd influx** — new people still arriving.
4. **Super-exponential growth** — attention whose *growth rate* is itself growing.

(4) is Sornette's LPPLS bubble signature applied to the *crowd* instead of the chart: attention spreads as an epidemic, and a contagion that accelerates must saturate.

Per-name ranking is the whole trick. “Extreme” always means extreme *for this name*, so BTC is judged against loud-for-BTC and a utility ETF against quiet-for-utilities. One global threshold would fire on BTC every week and never on anything else.

What never enters the prediction: the price chart. Price *gates* an ending (you cannot end a party that never started) and *grades* every alert — but the crowd does the predicting.

That restriction costs measured capture. It is kept because it is the only way to claim: *the crowd alone called the top*.

The evidence base

9 years

of daily Reddit attention,
2017–2026

333

hype episodes labelled
before any detector
existed

122

tops the detector could
possibly have caught

12,528

authors scored on their
own forecasting record

- **The exam was written before it was sat.** Episodes, tops and the detectable subset are defined in Notebook 01 by a rule applied to price alone — with no detector in existence and no feature consulted. Nothing downstream can move the target.
- **Two denominators are quoted everywhere.** 24 of 122 *detectable* tops is also 7.2% of all 333 episodes. Crowd coverage is deep in 2020–22 and 2026, thin to absent in 2023–25: there the detector is **blind**, which is not the same as wrong.
- **Walk-forward throughout, against real closes.** Prices are Bloomberg; every threshold is fitted on *earlier years only* and then frozen, and the live pipeline snapshots each day's signal and never revises it.

SO WHAT The honest headline is not a hit rate — it is that the target, the split and the budget were all fixed in advance, so the hit rate means something.

The strongest result: the danger state

62%

of danger-state days are followed by a $\geq 10\%$ -in-7-days drop within 30 days

19%

on ordinary candidate days

90% instrument-cluster CI on the uplift: [+28pp, +50pp]

The danger state is not a model. It is the conjunction of two things the system already computes:

- the crowd is $\geq 2\times$ its own 120-day normal, *and*
- price is already in a boom off its own 120-day low.

A combined price+crowd *alert* model was also tested and did **not** clear its criterion: its 65% hit rate is indistinguishable from the 62% baseline of its own candidate days.

Which is itself the finding: the *state* carries the information, so the product ships it as a standing amber band and uses the END alerts to time the peak *within* it.

SO WHAT The most useful thing this system produces is a *regime flag* that roughly triples the base rate of a near-term double-digit drawdown.

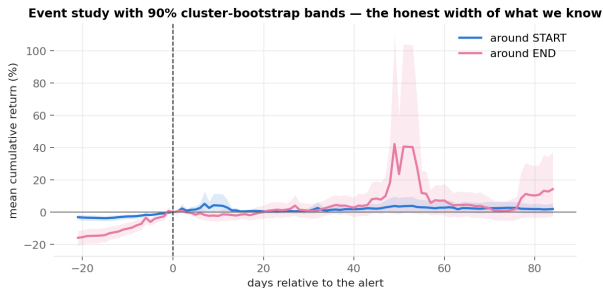
The walk-forward operating record

	Captured	Detectable	Rate	False alarms	FA / instr-yr
GET OUT (shipped: boom-gated, smoothed)	24	122	19.7%	39	0.195
GET IN (shipped: phase-aware, smoothed)	20	125	16.0%	124	0.25

- **Lead time.** GET OUT fires a median **8 days before the peak**; GET IN a median 17 days after the trough, with a median **66 days of rally still ahead**.
- **Read those two differently.** 66 days of rally ahead is *actionable*. Eight days before a top is not time to build a short — it is time to *stop adding*. The deck says so rather than rounding it up into an entry signal.
- **One number is above budget, and it is printed on the pane itself:** GET IN false alarms run 0.25/instrument-year against a derived budget of 0.23.

SO WHAT One in five detectable tops, at one false alarm per five instrument-years, with eight days' warning. Useful as a risk flag; nowhere near a standalone strategy — and priced as such.

What actually happens after a GET OUT



Median forward path of every scored GET OUT, one vote per instrument, with 90% cluster-bootstrap bands. Red *rising* means price *falling*: the axis is drawn so that “the signal was right” always points up.

SO WHAT Ten trading days after a GET OUT the average name has done **−2.6%** against **+0.6%** on a random day — 90% interval $[-6.6\%, -0.3\%]$, which *excludes zero*. It is the only horizon of four tested (3 / 10 / 21 / 84 days) that does.

The desk complaint, fixed at the signal and not with a rule

The complaint: “euphoria for a single day, then gone” — and START markers landing next to END markers.

The tempting fix: a minimum-duration rule. Rejected — it hides the symptom and adds a constant defensible only by taste.

What was done instead: two changes to the signal, each measured.

- Trigger on the **7-day-smoothed** score. Quality AP 0.435 → 0.449, at a recorded cost of 2 captures.
- Make GET IN **phase-aware**: a day that already meets every ending condition is end-stage, so a “start” there cannot fire.

20 → 2

START markers printing adjacent to an END, before and after

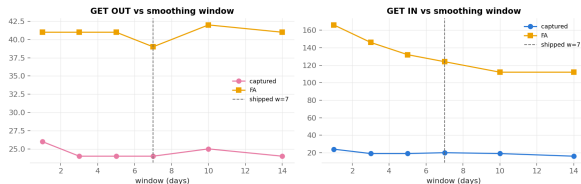
29 → 20

captures — the price paid, recorded rather than buried

Why pay it: a START landing on an END destroys trust in the whole pane, and an untrusted pane has an effective hit rate of zero.

SO WHAT This is the methodological claim of the project in one slide: fix the *signal*, publish what the fix cost, and never patch the output with a rule you cannot derive.

The gate that *added* hits by firing less often



GET OUT quality against the smoothing window. Seven days is a plateau, not a spike — which is what a real effect looks like and what a fitted constant does not.

SO WHAT Requiring an *actual* price boom before an ending can be called took captures from **16 to 26** of 122. Fewer alerts, *more* hits: the gate stopped the detector spending alerts on names that were never in a rally. The boom thresholds are not new constants — they are the same ones the ground-truth labels already use.

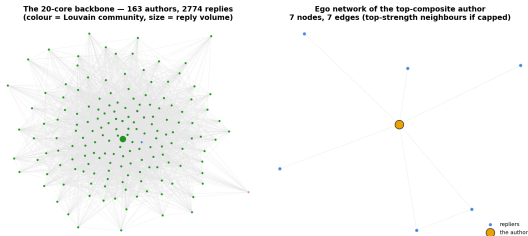
What was tested and rejected

Claim	n	Alert 20d	Baseline	Diff	90% CI
GET IN → BUY	215	+0.34%	+0.98%	−0.63%	[−1.51%, +0.26%]
GET OUT → SELL	67	−0.21%	+1.34%	−1.55%	[−5.63%, +2.63%]

- **Criterion, written before any return was computed:** $n \geq 30$ judged alerts; mean 20-day forward return beats the same-instrument unconditional baseline in the expected direction; 90% instrument-cluster CI of the difference excludes zero.
- **Both REJECTED.** The SELL direction is right, but at $n = 67$ the interval spans zero. The BUY edge is simply not there at 20 days. A short-after-GET-OUT overlay was also measured: annualised Sharpe -0.82 .
- **A gradient-boosted model** on the same features caught 11 endings against the shipped rules' 23, inside the same tournament — and the weights it *learned* rank the same features top, which is independent evidence the rules are not arbitrary.

SO WHAT Avoiding the left tail of a long you already hold is a different and easier problem than timing a short entry. The evidence supports the first claim only, so that is the only claim made.

Whose voice is worth anything? (Chan 2026, replicated)



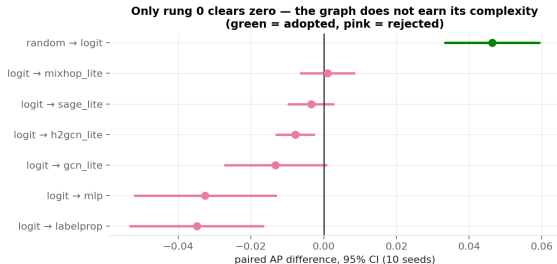
The 20-core backbone of the reply graph: 163 authors, 2,774 replies. Position is force-layout, size is how many people reply to them, colour is measured usefulness.

SO WHAT Every author's calls are scored against the next 20 days, on a bar that *scales with the name* (3% is a real call on an index and noise on a meme stock), then shrunk toward the crowd base rate so nobody tops the board on three lucky calls.

The counter-intuitive finding, reproduced:

the accounts with the most reach were among the *least* accurate — $3\times$ the degree, barely-above-chance hit rate. The list a follow-the-big-names desk would copy is flagged in the product as the list to **fade**.

The influence result we are *not* allowed to claim



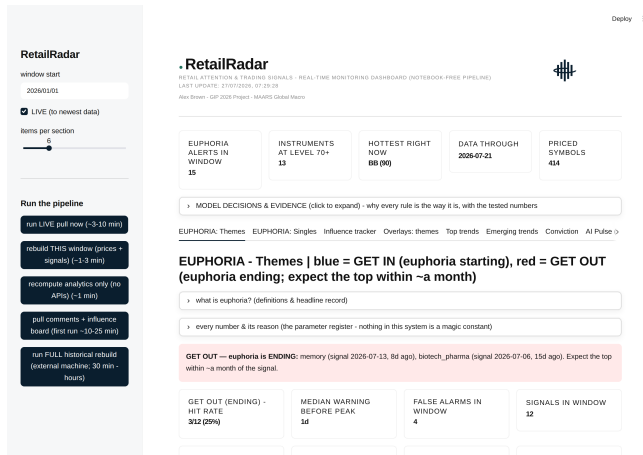
The adoption ladder. Each rung is a more complex model against the one below it: 10 paired seeds, cluster-bootstrapped 90% interval on the AP difference, Bonferroni-corrected across six comparisons.

SO WHAT Eight graph architectures were tested — GraphSAGE, GCN, SGC/MixHop, H2GCN, label propagation, an MLP. **Only rung 0 clears zero:** a plain linear model beats random, and *nothing beats the plain linear model*.

Our AP of **0.098** against a 0.047 random floor is in the same place as the thesis's own GraphSAGE result (0.140 on its data), so this is a replication of a small effect — not a failure to find a large one.

So the product ships no model here. It shows the measured record only.

The product: two lines, and nothing else to interpret



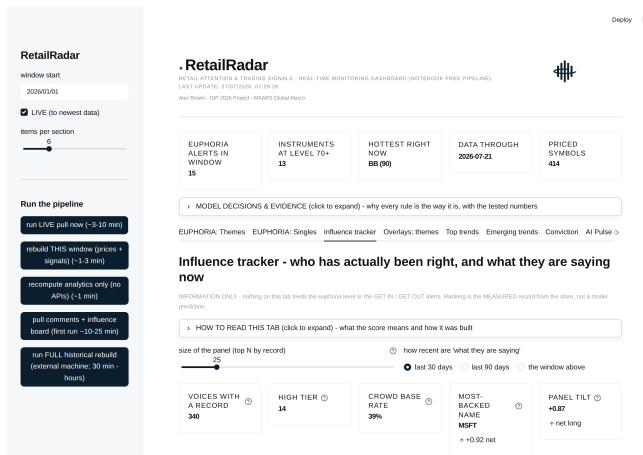
The themes pane, as shipped.

SO WHAT Blue = GET IN, red = GET OUT. Nothing else on the chart is a decision; the 0–100 curve is context.

The panes are ordered by *most recent signal*, so the top of the screen is the part of the book that changed.

Every explanation is plain English with the measured number attached, read live from the notebook exports — so nothing on screen goes stale.

The product: what the names with a record are pushing



The influence pane, as shipped.

SO WHAT Bubble size is how much a voice is worth, colour is which way it leans. Below it: the leaderboard, the reply map, and the “loud but wrong” list.

This pane does not feed the euphoria signal. It is background colour — *“the accounts with a track record are leaning short semis this month”* — and it is labelled as such on the screen.

The store is pseudonymous and text-free by a hard write-time check — by construction, not by convention.

Where this stands, and what is next

What is defensible today

- A regime flag that triples the base rate of a near-term 10% drawdown.
- An ending signal with 8 days' median warning at one false alarm per five instrument-years.
- A register in which every constant is learned, derived, a labelled convention, or a named desk decision.
- Two candidate trades and eight graph models *rejected* on pre-stated criteria.

The single biggest available gain

- **More crowd, not more model.** The comment backfill is $\sim 10\times$ the post volume and feeds every one of the four ingredients. Coverage — not architecture — is what makes 2023–25 blind.
- Then, and only then, re-run the same adoption ladder. Nothing is adopted without a paired-seed interval that clears zero.

Full evidence: 77-page deck, seven notebooks,
`PARAMETER_REGISTER.md`, `DECISIONS.xlsx`.

SO WHAT The result worth defending is not the hit rate — it is that every number here was fixed in advance, every rejection was published, and the dashboard shows only what survived.